



Crypto AI Forecast Dashboard

An end-to-end Machine Learning and NLP platform designed to analyze market sentiment and forecast cryptocurrency price trends. This project integrates statistical models, deep learning, and real-time news analysis into a unified Streamlit interface.



Features

- **Hybrid Sentiment Engine:** A dual-layered approach using **VADER** and **FinBERT** to decode market psychology from live news headlines.
- **Multimodal Forecasting:**
 - **LSTM (Deep Learning):** Long Short-Term Memory networks for complex pattern recognition.
 - **FB Prophet:** Robust time-series forecasting focused on crypto-market seasonality.
 - **ARIMA:** Classic statistical modeling for linear trend analysis.
- **Live Market Marquee:** A non-blocking, asynchronous "marquee" built with Streamlit Fragments that streams live sentiment across Bitcoin, Ethereum, and Solana.
- **Interactive UI:** A fully responsive dashboard with real-time clock synchronization and isolated component rerendering for high performance.
- **Secure Authentication:** Integrated with **Supabase Auth** for personalized user sessions and secure data access.



Project Structure

The project follows a modular architecture to ensure scalability and clean code separation:

```
crypto-forecaster/
├── app.py                # Main Dashboard (Streamlit)
├── src/
│   ├── sentiment/
│   │   ├── analyzer.py   # Hybrid sentiment processing logic
│   │   └── utils.py      # Shared categorization & map functions
│   ├── models/
│   │   └── sentiment_logic.pkl # Trained transformer/VADER package
│   └── processing/
│       └── data_loader.py # News & Price data ingestion
├── Output/
│   ├── Arima/            # Exported forecast plots
│   ├── FB_Prophet/
│   └── LSTM/
└── requirements.txt      # Project dependencies
```



Model Performance

Component	Metric	Score
Sentiment Analysis	Macro F1-Score	0.80

Component	Metric	Score
Fear Detection	Recall	0.94
Price Forecasting	Horizon	30 Days

Getting Started

Installation

1. **Clone the repository:**

```
git clone https://github.com/yourusername/crypto-forecaster.git
cd crypto-forecaster
```

2. **Install dependencies:**

```
pip install -r requirements.txt
```

3. **Launch the Dashboard:**

```
streamlit run app.py
```

Tech Stack

- **NLP:** Hugging Face Transformers, NLTK, VADER
- **ML/DL:** TensorFlow, Keras, Scikit-Learn, FBProphet, Statsmodels
- **Frontend:** Streamlit
- **Visualization:** Matplotlib, Plotly

License

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Contributors

This project was a collaborative effort by the following developers:

- **Rushik Kokate (@Rushik): Project Lead & Lead Developer.** Architected the Hybrid Sentiment Engine, implemented the fragmented Streamlit UI, and refined the core data processing pipelines.

- **Shravan Shidruk (@Shravan): ML Engineer & Dashboard Developer.** Developed the Analytics and Dashboard components. Specifically responsible for developing and training the **ARIMA**, **FB_Prophet**, and **LSTM** models for the following tickers:
 - **BTC** (Bitcoin)
 - **ETH** (Ethereum)
 - **SOL** (Solana)
 - **USDC** (USD Coin)
 - **USDT** (Tether)
 - **Stefie George (@Stefie): Market Sentiment Model Developer.** Designed the initial sentiment analysis prototype and established the text processing foundations in the early project phases.
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Changelog Report (Group 7 , 4.2 Batch)

All notable changes to this project will be documented in this file.

[2026-02-11] - Sentiment Engine & UI Integration

- **Merged** branch `shravanFeature` into main, integrating the initial Streamlit `app.py` dashboard framework. (**Contributor: Rushik**)
- **Added** hybrid VADER/FinBERT sentiment logic with global scope functions for successful pickling. (**Contributor: Rushik**)
- **Improved** Enhanced `map_emotion` keywords to include crypto-specific signals like "crash" and "rally". (**Contributor: Rushik**)
- **Fixed** Resolved `EOFError: Ran out of input` by ensuring atomic writes and `KeyError: 'text'` by targeting the `title` column. (**Contributor: Rushik**)

[2026-02-10] - Analytics & Dashboard Merge

- **Added** `src/sentiment/analyzer.py` to provide a modularized interface for the Streamlit dashboard to calculate real-time market sentiment percentages. (**Contributor: Rushik**)
- **Added** `src/sentiment/models/sentiment_logic.pkl` containing the trained VADER analyzer and emotion mapping logic. (**Contributor: Rushik**)
- **Added** `src/sentiment/Train_sentiment_.py` to implement hybrid VADER/FinBERT sentiment logic with global scope functions for successful pickling. (**Contributor: Rushik**)
- **Added** `src/sentiment/test_sentiment.py` to validate model accuracy against GitHub news data and generate market psychology distributions. (**Contributor: Rushik**)
- **Refactored** original NLP logic from `NLP.py` into a modularized system: `Train_sentiment_.py`, `test_sentiment.py`, and `analyzer.py`. (**Contributor: Rushik**)
- **Added** ARIMA forecasting results for multiple coins including BTC, ETH, SOL, USDC, and USDT. (**Contributor: Shravan Shidruk**)
- **Added** Walk-forward validation plots and overview visualizations in `Output/Arima/`. (**Contributor: Shravan Shidruk**)
- **Added** `notebooks/USDT_INR/first_attempt.ipynb` for initial Tether price analysis. (**Contributor: Shravan Shidruk**)
- **Changed** `requirements.txt` updated to track new dependencies for data science modules. (**Contributor: Shravan Shidruk**)

[2026-02-04] - Initial Sentiment Logic (nlp branch)

- **Added** `NLP.py` containing the initial prototype for sentiment analysis and text processing. (**Contributor: stefiegeorge**)
- **Created** Logic foundations for categorizing financial headlines which served as the basis for the current sentiment pipeline. (**Contributor: stefiegeorge**)

[2026-02-02] - Data Pipeline Refactoring

- **Added** `src/api/fetch_news.py` for collecting cleaned bitcoin news. (**Contributor: Rushik**)

[2026-01-21] - Path & Performance Optimization

- **Added** `src/config.py` for centralized path management. (**Contributor: Rushik**)
- **Fixed** Resolved `YFRateLimitError` via batch downloads in `fetch_data.py`. (**Contributor: Rushik**)
- **Changed** `clean_data.py` now converts prices to INR using live exchange rates. (**Contributor: Rushik**)

[2026-01-20] - Initial Data Setup

- **Added** Basic `fetch_data.py` script. (**Contributor: Rushik**)
- **Created** Folder structure for `data/raw` and `data/processed`. (**Contributor: Rushik**)